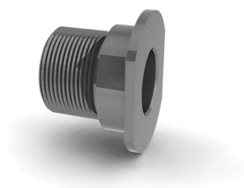
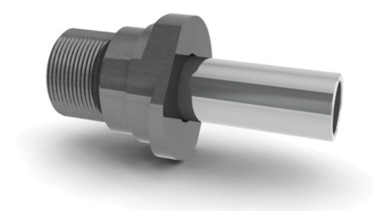
Drill Bushings

THE HARCOURT AIRFEED SYSTEM

Harcourt Airfeed drill bushing tips provide a fast and accurate method for portable or stationary mounting of drilling, tapping, and other self-contained machining units on a jig or fixture. The threaded end of the drill bushing tip screws into the nose end of the drilling unit. The helical flanges on the tip fit under the shoulders of the helix lock liner bushing, airfeed lockscrews, or lock strips on the jig, effectively holding the drilling unit in alignment position and absorbing thrust and torque.

Mounting Information:





Nosepieces

Automatic, mechanical, or airfeed drills, tappers, and back spot-facers are equipped with either a short or a long nosepiece depending on the length of the cutting tool holding device - i.e. drill chuck, tap chuck, Morse taper adapter, ect. For detailed information contact Harcourt.

Reducer Bushings

Reducer bushings are occasionally used to reduce the thread size in a specific nose piece to adapt it to accommodate smaller thread drill bushing tips. For complete dimensional information refer to pages 203.

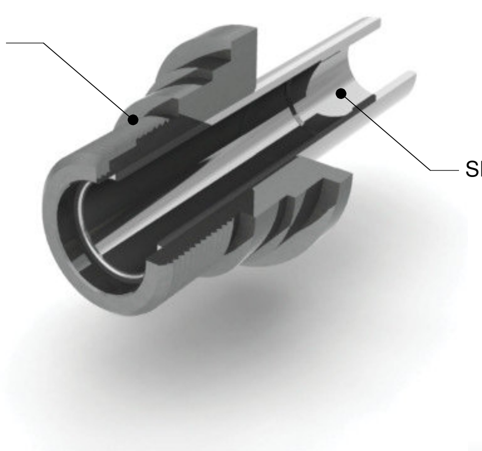
Drill Bushing Tips

The drill bushing is shown in position for insertion into the four most frequently used locking devices. Close fit between mating shank of tip and hole in lock liner bushing holds tool in alignment to produce straight close-tolerance holes. See pages 198-199.

Construction:

All drill bushing tips are constructed by a two-piece method consisting of a collar and a pressed-in shank. The collar contains the screw threads, alignment diameter, and the lock flange. The shank contains the A, B, and C dimensions listed in the chart on pages 198-199.
Maximum and minimum drill sizes (A) for drill bushing tips are shown in the column headings of the chart on pages 198-199.

Collars or shanks only of two-piece drill bushing tips may be ordered separately for replacement purposes. (See 'Ordering information' on the following page.)



THE HARCOURT AIRFEED SYSTEM

General Information:

Standard ID Tolerances:

IMPERIAL

FROM #70 - 1/4	+.0001 / +.0004
OVER 1/4 - 3/4	+.0001 / +.0005
OVER 3/4 - 1-1/2	+.0002 / +.0006

METRIC

FROM 1 - 3mm	+.002 / +.008
OVER 3 - 6mm	+.004 / +.012
OVER 6 - 10mm	+.005 / +.014
OVER 10 - 18mm	+.006 / +.017
OVER 18 - 30mm	+.007 / +.020

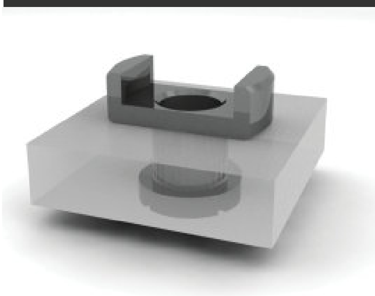
Harcourt Airfeed drill bushing tips are finished to order to the drill size specified by the user. They are available for applications from #70 drill up to through 1-15/32" diameter.
Harcourt Airfeed drill bushing tips and accessories fit all standard automatic feed drill motors including Quackenbush, Gardner-Denver, Aro, Keller, Buckeye, Ingersoll-Rand, etc.

For mounting dimensions for lockscrews and lock strips see page 204.

Carbide Shanks:

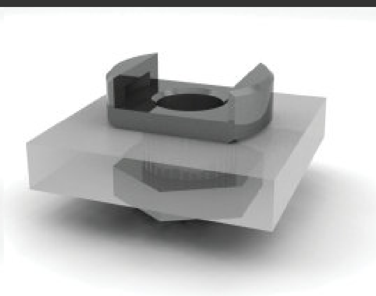
Harcourt Airfeed drill bushing tips are also available with carbide shanks to minimize wear on the bushing and maximize the drill bushing's life. Carbide shanks are used especially when drilling in high temperature and/or high volume applications.
To order carbide shanks specify the drill bushing tip as 'AC-#####'. For more ordering information look below in 'Ordering Information'.

Mounting Options:



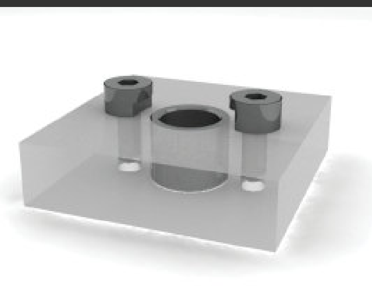
Helix lock liner bushings and lock nut or lock ring.

The most popular and preferred mounting method has a lock liner with a helix angle cut into the i's ears to ensure a 100% positive lock. The angle produces a lock such that vibrations, torque, and, thrust are greatly reduced. Lock liners are doweled in place to aviod twisting upon use.



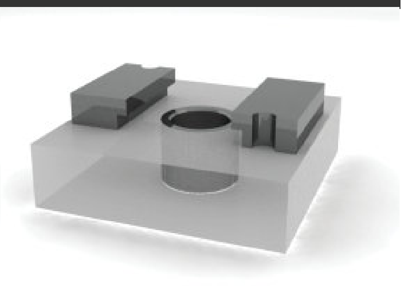
Lock liner bushings and lock nut or lock ring.

Conventional method for mounting automatic, mechanical, or airfeed drills, tappers and spot-facers to a jig or fixture. A hole is bored in the jig to accommodate the lock liner bushing. The lock ring or lock nut holds the lock liner bushing in place.



Airfeed lockscrews and liner bushings

An alternate mounting method for holes so closely spaced that lock liner bushing cannot be used has lockscrews mounted directly into the jig. The shank of the drill bushing tip fits directly into a bored hole in the jig. Hardened press fit liner bushings are available to press into the bored hole to provide a wearing surface and prolong the useful life of the jig.



lock strips and liner bushings

A second mounting method for closely spaced holes employs a lock strip along each side of a row of holes in the jig. The flanges on the drill bushing tip lock under the extended edges of the lock strip when the tip is turned 30° counter-clockwise. Press fit liner bushings pressed into the bored holes in the jig are recommended to forestall wear in the bored hole.

Ordering Information:

Drill Bushing Tips:

Specify part number selected and name; add drill size as a suffix to part number. Drill size can be specified in either a decimal or fraction.
There is an extra charge for all drill bushing tips specified to have smaller than the minimum drill size. Standard drill bushing tips are those with standard drill sizes, and with sizes falling between the maximum and minimum shown in row 'A' on the following page.

Shank Only:

For all drill bushing tips with a prefix 'A', select the two-piece tip of the proper size and specify this number, omitting the prefix 'A'. Specify the part name. Add drill size as a suffix to the part number by specifying it as a fraction or decimal to four places.

Collar Only:

Specify Collar part number shown directly below Collar drawing at head of columns.

Ordering Examples:

Complete adapter tip assembly:
A-21101-.1250
Shank only:
21101-.1250
Collar only:
21000
Coolant type assembly:
AU-21101-.1250
Complete assy. w/ carbide shank:
AC-21101-.1250